Computer Science BSc; Analysis-2
The theorems to be proved in the Advanced Level Part of the Theoretical Exam
(26 theorems)
2024/2025 Autumn

The abbreviation "AN-2" denotes the following reference:
Analysis 2 lecture schemes, written by István Csörgő

The abbreviation "AN-3" denotes the following reference:
Analysis 3 lecture schemes, written by István Csörgő

They can be found in the Digital Library of the Faculty of Informatics or they can be opened from CANVAS

1. The theorem about the connection between the differentiability and the continuity
   ("AN-2 " p. 17)

2. The derivatives of the following functions: c, $ax+b$, x^n, e^x, $\sin x$
   ("AN-2 " pp. 17-18)

3. The derivative of the sum
   ("AN-2 " p. 18)

4. The derivative of the product
   ("AN-2 " pp. 18-19)

5. The derivatives of $\tan x$, $\ln x$
   ("AN-2 " p. 20)

6. The derivative of arcsin and of arctan
   ("AN-2 " pp. 26-27 and lecture)

7. The First Derivative Test (First Order Necessary Condition) for local extremum
   ("AN-2 " p. 22)

8. Rolle's Mean Value Theorem
   ("AN-2 " p. 23)

9. Cauchy's Mean Value Theorem
   ("AN-2 " p. 23)

10. Lagrange's Mean Value Theorem
    ("AN-2 " p. 23)

11. The First Derivative Test for monotonicity
    ("AN-2 " p. 24)

12. L'Hospital's Rule (the case that was proved on the lecture)
("AN-2 " p. 28)

13. The connection between the derivatives of a function and of its Taylor-polynomials. Prove it for the 0-th and for the 1-st derivatives.
("AN-2 " p. 29 and lecture)

14. The theorem about the Taylor-formula
("AN-2 " p. 30 and lecture)

15. The theorem about the set of all antiderivatives of a function
("AN-2 " p. 33)

16. The 5 simple integration rules
("AN-2 " pp. 34-35)

17. The rule of Integration by Parts for antiderivatives
("AN-2 " p. 38)

18. The rule of Substitution Form I. for antiderivatives
("AN-2 " p. 38)

19. Prove by definition that the constant function is Riemann-integrable
("AN-2 " p. 41)

20. The theorem about the connection between the integrability and the oscillation sums
("AN-2 " p. 42)

21. The Fundamental Theorem of Calculus (the Newton-Leibniz Formula)
("AN-2 " p. 51)

22. The rule of Integration by Parts for definite integrals
("AN-2 " p. 52)

23. The theorem about the integrability of the continuous functions
("AN-2 " p. 47)

24. The theorem about the continuity of the Integral Function
("AN-2 " p. 49)

25. The Newton-Leibniz formula for improper integrals
("AN-2 " p. 56)

26. The First Derivative Test (First Order Necessary Condition) of local extremum for functions of type $\mathbb{R}^n \to \mathbb{R}$
("AN-3 " p. 41)